

COSMIC Measurement Manual

Part 2: Guidelines

1. Identification of Functional Users

GUIDANCE on Rule 7: Identification of the functional users

The identification of functional user (or users) is determined by the Functional “User” Requirements that must be measured and by the purpose of the measurement.

NOTE: *There is nothing absolute about a functional user, i.e. identify the functional users per functional process. A sender/receiver of data may be a functional user of one or more functional processes, but not a functional user of other functional processes, even in the same software being measured.*

2. Identifying Functional Processes

The first step of the Measurement Phase is to identify the set of functional processes of the piece of software to be measured, from its FUR.

The relationships between a triggering event, the functional user and the Entry data movement that triggers a functional process being measured are presented in Figure 3.2 where: a triggering event causes a functional user to generate a data group that is moved by the triggering Entry of a functional process to start the functional process.

NOTE: *For ease of reading, the reference to the data group is omitted when stating that a functional user initiates a triggering Entry that starts a functional process, or even more simply that a functional user initiates a functional process.*

Guidance on Rule 10: Identification of Functional Processes

The process of identifying functional processes, after the functional users have been identified, given the FUR for the software being measured follows the chain of Figure 3.2:

- 1 Identify the separate events in the world of the functional users that the software being measured must respond to - the “triggering events”.
- 2 Identify which functional user(s) of the software may respond to each triggering event.
- 3 Identify the data group(s) (i.e. the triggering Entry or Entries) that each functional user may initiate in response to the event.
- 4 Identify the functional process started by each triggering Entry.

NOTE: Triggering events can be identified in state diagrams and in entity life-cycle diagrams, since some state transitions and entity life-cycle transitions correspond to triggering events to which the software must react.

3. Identification of Data Groups

Having identified the functional processes, the next step is to identify their data movements. The following guidance assists in the identification of data groups and hence objects of interest particularly in the output of functional processes.

GUIDANCE on Rule 11: Identifying different data groups moved in the same one functional process

For all the data attributes appearing in an Entry-Exit-Read-Write of a functional process:

- a** Sets of data attributes that have different frequencies of occurrence describe different objects of interest.
- b** Sets of data attributes that have the same frequency of occurrence but different identifying key attribute(s) describe different objects of interest.
- c** All the data attributes in a set resulting from applying parts a) and b) of this guidance belong to the same one data group, unless the FUR specify that there may be more than one data group describing the same object of interest (see the Guidance on Rules 13 and 14 - Data Movements Uniqueness, cases b) and c)).

NOTE: A functional user of the software being measured may be the object of interest of a data group sent or received by the functional user.

NOTE: In theory, a data group might contain only one data attribute if this is all that is required, from the perspective of the FURs, to describe the object of interest. In practice, such cases occur commonly in real-time application software (e.g. the data group entered to convey the tick of a real-time clock or the entry of the state of a sensor); they are less common in business application software.

NOTE: There is nothing absolute about an object of interest, i.e. identify the objects of interest per functional process. A “thing” may be an object “of interest” to a functional user via one or more functional processes, but not be an object “of interest” to another functional user via other functional processes, even in the same software being measured.

The origin of a data group can be of many forms, e.g.:

- a** A physical record structure on a hardware storage device (file, database table, ROM memory, etc.).
- b** A physical structure within the computer’s volatile memory (data structure allocated dynamically or through a pre-allocated block of memory space).
- c** A clustered presentation of functionally-related data attributes on an input/output device (display screen, printed report, control panel display, etc.).
- d** A message in transmission between a device and a computer, or over a network, etc.

3.1 About the Identification of Objects of Interest and Data Groups

The definition and principles of objects of interest and of data groups are intentionally broad in order to be applicable to the widest possible range of software: this sometimes results in it being difficult to apply the definition and principles when measuring a specific piece of software. See Part 3 for examples to assist in the application of the principles to specific cases.

When faced with a need to analyze a group of data attributes that is moved in or out of a functional process or is moved by a functional process to or from persistent storage, it is critically important to decide if the

attributes all convey data about a single “object of interest”, since it is the latter that determine the number of separate “data groups” as defined by the COSMIC method that will be moved by data movements.

For instance, if the data attributes to be input to a functional process are attributes of three separate objects of interest, then three separate Entry data movements must be identified.

Deciding on the number of data groups can be difficult when analyzing the output of a functional process of a business application which may include:

- Multiple data groups, each describing a different object of interest, e.g. a report showing totals at various levels of aggregation.
- The results of enquiries where the output will vary depending on the input.
- Data groups that may even be unrelated to each other, e.g. an invoice which includes an advertisement for an unrelated service.

When analyzing complex output, e.g. reports with data describing several objects of interest, consider each separate candidate data group as if it were output by one separate functional process. Each of the data group types identified this way must also be distinguished and counted when measuring the complex report.

3.2 Data or Groups of Data That Are Not Candidates for Data Groups

Any data appearing on input or output screens or reports that are not related to an object of interest to a functional user should not be identified as indicating a data group.

The COSMIC Generic Software Model assumes that all manipulation of data within a functional process is associated with the four data movement types. Hence no data groups may be identified arising from data manipulation within a functional process in addition to the data groups moved by the Entries, Exits, Reads and Writes of the functional process.

3.3 Identification of Data Attributes (Optional)

In the COSMIC method, it is not mandatory to identify the data attributes. However, understanding the concept of a “data attribute” is necessary to understand how to measure changes. A requirement to change a data attribute can result in the data movement to which the attribute belongs being indicated as “changed”.

Also, it may be helpful to analyze and identify data attributes in the process of distinguishing data groups and objects of interest.

4. Identification of Data Movements

GUIDANCE on Rule 16: Entry (E)

- a The data group of a triggering Entry may consist of only one data attribute which simply informs the software that “an event Y has occurred”.
- b For clock-ticks that are triggering events identify an Entry from a functional user, in this case the Clock.
- c Unless a specific functional process is necessary, obtaining the date and/or time from the system’s clock is not considered an Entry or any other COSMIC data movement.

NOTE: Very often, especially in business application software, the data group of the triggering Entry has several data attributes which inform the software that “an event Y has occurred and here is the data about that particular event”.

GUIDANCE on Rule 17: Exit (X)

- a For an enquiry which outputs fixed text, where “fixed” means the message contains no variable data values, identify an Exit for the fixed text output.
- b When identifying Exits, ignore all fields and other headings that enable human users to understand the output data.

GUIDANCE on Rule 18: Read (R)

Do not identify a Read when the FUR of the software being measured specify any software or hardware functional user as the source of a data group, or as the means of moving the data group to persistent storage. Interaction with other functional users is by definition across a boundary (Generic Software Model, principle 1), which is handled by an Entry data movement. The actual Read takes place within the boundary of the software being measured.

GUIDANCE on Rule 19: Write (W)

Do not identify a Write when the FUR of the software being measured specify any software or hardware functional user as the destination of the data group or as the means of retrieving a persistently-stored data group. Interaction with other functional users is by definition across a boundary (Generic Software Model, principle 1), which is handled by an eXit data movement. The actual Write takes place within the boundary of the software being measured.

NOTE: When the FUR require data to be stored or to be retrieved from storage, the measurer must investigate whether the data can be stored or retrieved within its own boundary, i.e. to/from “persistent storage”, or whether data is required to be stored/retrieved with help of a functional user of the software being measured (i.e. via some other piece of software, or directly to or from a hardware device).

GUIDANCE on Rules 13 and 14 - Data Movements Uniqueness

- a Unless the FUR are as given in guidance b) or c), all data describing any one object of interest that is required to be entered into one functional process is identified as one data group moved by one Entry.
- b If FUR specify that different data groups must be entered into one functional process, each from a different functional user, where each data group describes the same object of interest, then one Entry is identified for each of these different data groups.
- c If FUR specify that different data groups must be moved from persistent storage into one functional process, each describing the same object of interest, then one Read is identified for each of these different data groups.

NOTE: A functional process may, of course, have multiple Entries, each moving data describing a different object of interest.

NOTE: The same equivalent guidance applies to any Read, Write or Exit data movement in any one functional process.

NOTE: The same equivalent guidance applies for Exits of data to different functional users from any one functional process.

NOTE: Any one functional process has only one triggering Entry.

NOTE: The same equivalent guidance applies for Writes in any given functional process.

NOTE: This guidance is analogous to rule b). In the case of the FUR to read different data groups describing the same object of interest, they will likely have originated from different functional users. In the case of the FUR to write different data groups, they will likely be made available to be read by different functional users.

GUIDANCE on Rules 16-17: Functional Process Requiring Data from a Functional User

- a When the functional process does not need to tell the functional user what data to send, a single Entry is sufficient (per object of interest).

- b** When the functional process needs to tell the functional user what data to send, an Exit followed by an Entry are necessary.

GUIDANCE on Rules 16-19: Control Commands in Applications with a Human Interface

In an application with a human interface, control commands are ignored as they do not involve any movement of data about an object of interest.

Error and confirmation messages are specific forms of an Exit and the Rules governing identification apply.

GUIDANCE on Rules 16-19: Error/Confirmation Messages and Other Indications of Error Conditions

- a** One Exit is identified to account for all types of error/confirmation messages issued by any one functional process of the software being measured from all possible causes according to its FUR.
- b** If a message to a human functional user provides data in addition to confirming that entered data has been accepted, or that entered data is in error, then this additional data is identified as a separate data group moved by an Exit in the normal way.
- c** All other data, issued or received by the software being measured, to/from its hardware or software functional users should be analyzed according to the FUR as Exits or Entries respectively, according to the normal COSMIC rules, regardless of whether or not the data values indicate an error condition.
- d** Reads and Writes are considered to account for any associated reporting of error conditions. Therefore, no Entry to the functional process being measured is identified for any error indication received as a result of a Read or Write of persistent data.
- e** No Entry or Exit is identified for any message indicating an error condition that might be issued whilst using the software being measured but which is not required to be processed in any way by the FUR of that software, e.g. an error message issued by the operating system.